

PyMeet v1.0.0

Student Name: Himanshu Goel

PyMeet is a modern, full-stack video conferencing application that provides real-time, high-quality audio, video, and text communication.

Features

- **Secure Authentication:** Register and login securely using JWT and bcrypt password hashing.
- **Instant Meetings:** Generate random secure links for immediate collaboration.
- **HD Video & Audio:** Powered by WebRTC protocols for minimal latency.
- **Screen Sharing & Chat:** Built-in tools for effective remote presentations.
- **PWA Ready:** Install the app directly on your device.

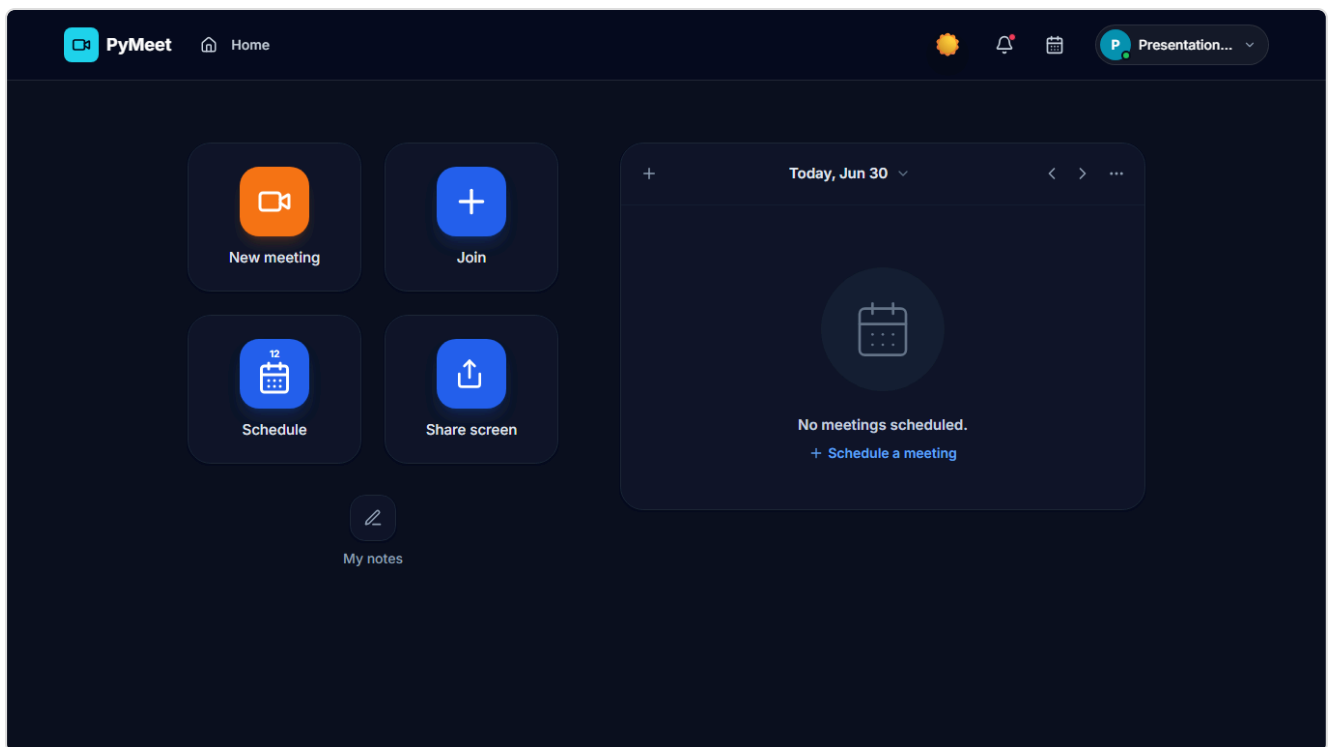
Tech Stack

- **Frontend:** React, TypeScript, Tailwind CSS
- **Backend:** Python, FastAPI, Uvicorn
- **Database:** Remote PostgreSQL Server (Live IP: 51.89.97.76)
- **Media Engine:** ZegoCloud API / WebRTC

Interface Preview

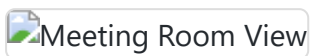
User Dashboard

A clean, responsive dashboard to manage your meetings.



Meeting Room

The core interface where users interact via video and chat.



How It Works

When a user registers, their details are securely hashed and sent to the remote PostgreSQL database. Upon login, the FastAPI server issues a JWT token. This token validates the user across the application. When joining a meeting, the frontend establishes a secure P2P WebRTC connection with other participants, bypassing the main server for media streams to ensure high performance.

Conclusion

PyMeet is a complete, production-ready solution that demonstrates the integration of complex backend data handling with real-time frontend media management.